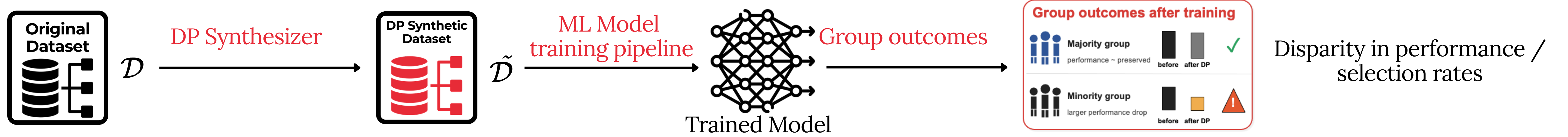




1. Introduction & Motivation

- **Synthetic data** is commonly discussed as a privacy-aware approach for data access and reuse [1]
- But **synthetic data is not private by default** [2], and information may still **leak**, as well as **reproduce & amplify built-in biases**

- **DP synthetic data generators** [3] address the privacy gap by producing a synthetic dataset with formal privacy guarantees
- Its noise **may distort subgroups' structure**, affecting **utility** and **fairness** (a.k.a "DP disparate impact") [4]



2. Research Questions

RQ1: Can fairness interventions recover DP-induced fairness loss?

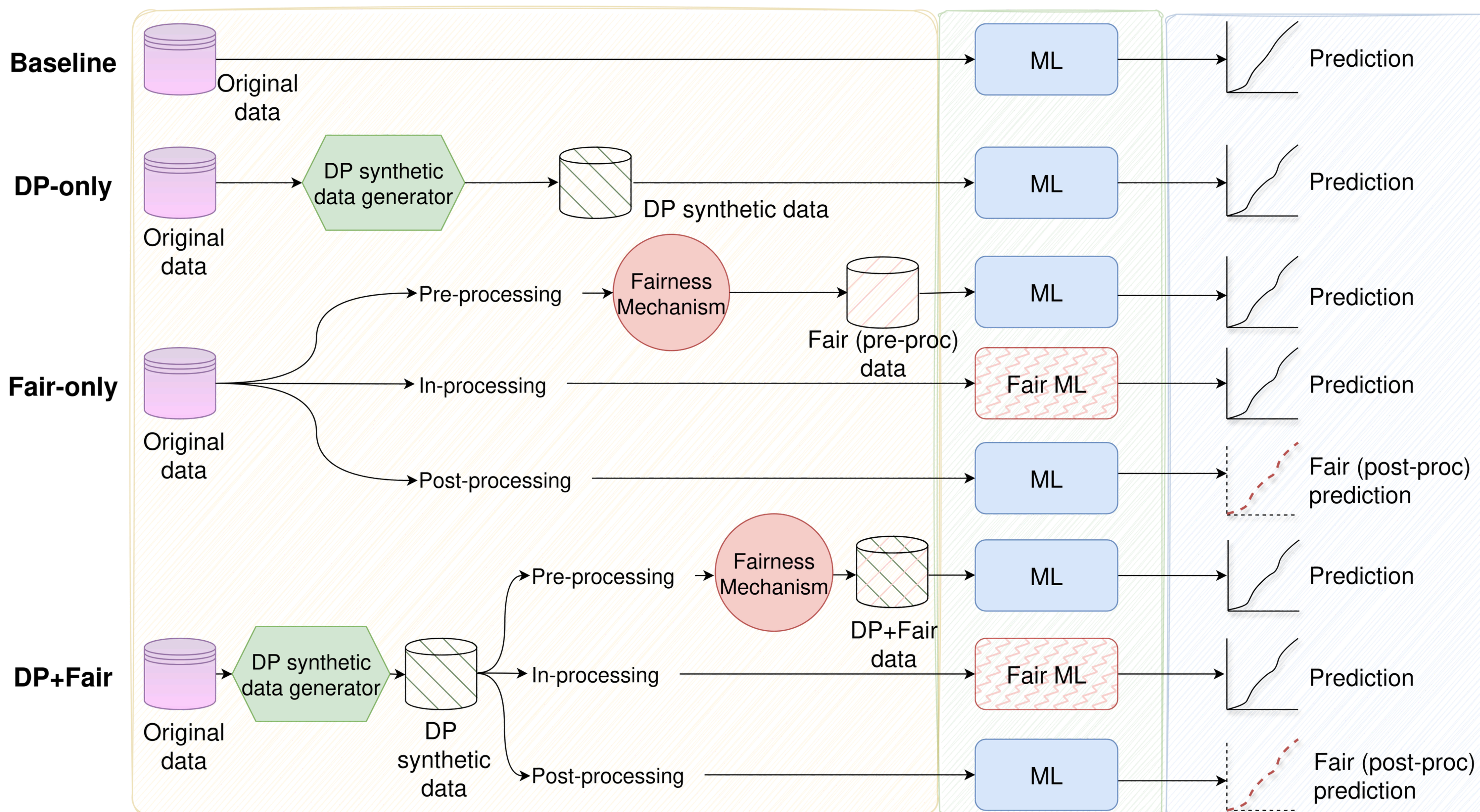
RQ2: Where should one intervene for the best privacy-fairness-utility trade-off

3. BenchmarkDPFair

Data pre-processing

Learning

Inference



4. Fairness Metrics

Metrics computation



Utility for group 1



Utility for group 2

$$\text{Metric} = \text{Utility for group 1} - \text{Utility for group 2}$$

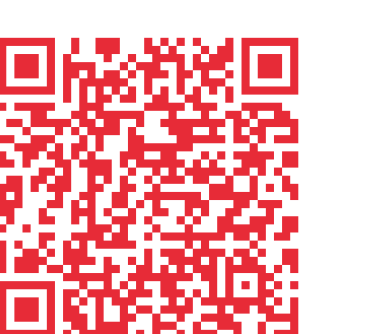
Metric	Utility
MAD	Accuracy
EOD	True Positive Rate
SPD	Positive Prediction Rate

Framework Available

`pip install BenchmarkDPFair`



PyPI

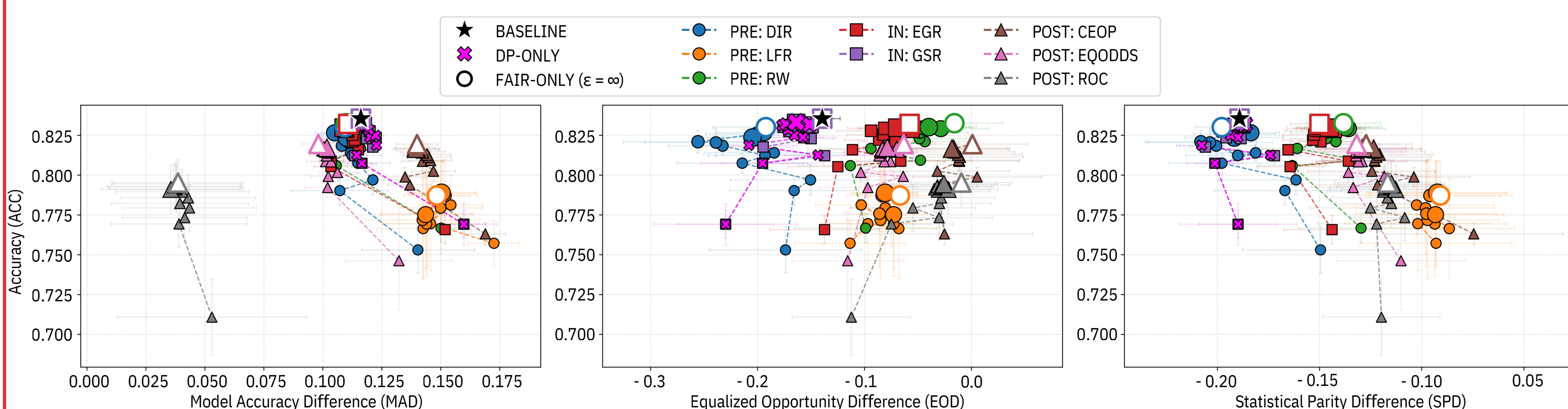


Code

5. Experimental Setup

Seeds	DP Synthesizers	Privacy budgets (ϵ)	Classifiers	Datasets	Ablation
20	2 marginal-based (AIM, MST)	12 $\rightarrow$ range (0.05 - 20)	• XGBoost • Logistic Regression • Random Forest	• 3 real-world • 6 synthetic with controlled bias	• DP-compliant calibration • In-processing hyperparameter tuning

6. Selected Result



Chosen setting: Adult · AIM · XGBoost

↑ Accuracy = better utility

→ MAD/EOD/SPD Toward 0 = better fairness

Larger marker = larger ϵ / weaker privacy

DP+Fair often moves closer to zero disparity than DP-only

Full benchmark on the paper, check it out!

7. Conclusion & Perspectives

Can we recover fairness?

- Yes — partially
- DP+Fair improves **EOD/SPD**
- Effects are metric-dependent

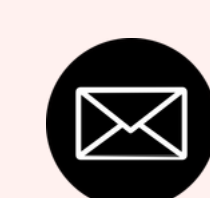
Where should we intervene?

- **Post-processing** gives the best fairness-privacy-utility trade-off

What remains open?

- Beyond **binary** tabular tasks • **Richer** protected attributes • **Joint** fair-private synthesizers

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- [2] Georgi Ganev and Emiliano De Cristofaro. 2025. The Inadequacy of Similarity-Based Privacy Metrics: Privacy Attacks Against “Truly Anonymous” Synthetic Datasets.
- [3] Ryan McKenna, Brett Mullins, Daniel Sheldon, and Gerome Miklau. 2022. AIM: an adaptive and iterative mechanism for differentially private synthetic data. VLDB 2022.
- [4] Georgi Ganev, Bristena Oprisanu, and Emiliano De Cristofaro. 2022. Robin hood and Matthew effects: Differential privacy has disparate impact on synthetic data. ICML 2022.